Jay Doshi

Jay.Doshi@rockets.utoledo.edu LinkedIn: jaydoshi1234 GitHub: jaydoshi2 +1 (419) 215-4693

Education

B.Sc. in Computer Science & Engineering **GPA: 3.7/4.0**
University of Toledo, Toledo, OH Expected Graduation: December 2027

Professional Projects

SkillBridge Learning Platform-AI | TF-IDF, NLP, Django, PostgreSQL, AWS **August 2024 – December 2024**

- Built an NLP-based recommender system using TF-IDF and cosine similarity, achieving 95% prediction accuracy in testing.
- Developed Gemini API-powered AI agent for auto-generating video transcripts and MCQs, reducing manual content creation effort.
- Designed PostgreSQL schemas with fact and dimensional modeling, improving query performance and reporting speed by 80%.

PolicyRAG Document Intelligence-AI | Python, FAISS, RAG, GPT-4o, FastAPI, Streamlit **December 2025 – January 2026**

- Indexed chunks in a FAISS vector store using OpenAI embeddings, reranking by cosine similarity to feed GPT-4o the top 5 clauses.
- Built a RAG pipeline over 5 insurance policy documents (500+ pages), extracting text and tables from PDFs using pdfplumber.
- Deployed a FastAPI endpoint with token auth answering batched queries in 40–50s; benchmarked retrieval latency and accuracy.

SportsSight Analytics-AI | Python, Computer Vision, YOLOv8, OpenCV, K-Means, NumPy **January 2025 – April 2025**

- Fine-tuned YOLOv8 on Roboflow football dataset, achieving 90% player and ball detection accuracy on test video clips.
- Applied K-Means pixel clustering for automated team assignment by t-shirt color, improving segmentation accuracy to 75%.
- Implemented optical flow and perspective transformation to compute real-world player speed and distance in meters.
- Generated real-time match analytics on ball possession and movement patterns to support data-driven sports insights.

DeepTruth-AI | Python, PyTorch, DistilBERT, Gemini API, GCP, AWS, MongoDB **January 2025 – March 2025**

- Fine-tuned DistilBERT on 9.5K+ labeled records, enabling dual-model scoring with 70% Gemini and 30% BERT weighting.
- Orchestrated scheduled Transformer retraining with AWS Lambda and CloudWatch cron jobs for ML lifecycle management.
- Automated CI/CD deployment via GitHub Actions and GCP VMs, reducing manual release steps and improving consistency.
- Won RocketHacks 2025 for developing scalable AI fact-checking product which processed data of 1000+ users during 6 months.

Experience

Network Engineer Intern | Cenovus Energy, Oregon, OH **September 2026 – Present**

- Monitor network and server health in SolarWinds NPM, flagging performance degradation before it escalates into outages.
- Collaborate with infrastructure, OT, and process-control engineers to validate system integrations and support production cutovers.

Data Processing Engineer Intern | Cleveland-Cliffs, Coatesville, PA **May 2026 – August 2026**

- Engineered a classification-based anomaly detection tool analyzing live gauge data, reducing defective steel plate escapes by 30%.
- Engineered sensor-data features via ETL pipelines from legacy OpenVMS Alpha into SQL Server, cutting manual extraction 40%.
- Translated production data into Power BI dashboards on SQL views and stored procedures, accelerating decisions by 50%.

IT Support Technician Level 2 | University of Toledo, Toledo, OH **September 2025 – May 2026**

- Resolved 40+ monthly IT tickets by troubleshooting Windows, hardware, software, and network issues, minimizing downtime.
- Managed 200+ IT assets through inventory tracking, device deployment, secure decommissioning, and life-cycle documentation.

Undergraduate Research Assistant | University of Toledo, Toledo, OH **May 2025 – December 2025**

- Designed and evaluated a PPO-based reinforcement learning model for autonomous drone target navigation using ROS2-Gazebo.
- Improved control-logic accuracy by 60% by computing RMSE and overshoot in Pandas and NumPy across 15+ hardware trials.
- Decreased target navigation error by 40% by tuning a custom PID-based flight model in a ROS2-Gazebo digital twin.

Technical Skills

AI & Machine Learning: Supervised/Unsupervised Learning, Deep Learning, Computer Vision, NLP, Reinforcement Learning, Transformer Models, Fine-Tuning, RAG, LLM, MLOps

Programming Languages & Frameworks: Python, Java, C++, C, SQL, R, JavaScript, FastAPI, Django

ML Libraries & Data: PyTorch, Hugging Face, FAISS, OpenCV, YOLOv8, Pandas, NumPy, Matplotlib, Power BI Dashboards

Databases & Cloud: PostgreSQL, MySQL, MongoDB, Firebase, AWS (Lambda, S3, EC2), Google Cloud Platform

Developer Tools: Git, GitHub Actions, Jupyter Notebook, Google Colab, VS Code, Claude-Code (plugins, skills, hooks)